

1 Confidence Interval Calculation

Table 1: Table 5.1 for Approx. Margin of Error (%)

Sample Size	Approx. Margin of Error (%) [*]
25	±22
50	±14
100	±10
250	±6
500	±4
1000	±3
1500	±2.5
2000	±2

^{*}The interval surrounding the actual sample outcome containing 95% of all possible sample outcomes.

1.1 Proportion of values (e.g. conf interval for correct predictions)

- Sample Proportion: $\hat{p} = \frac{n_{\text{True}}}{n}$
- Standard Error: $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
- Confidence Interval: $CI = \hat{p} \pm z \cdot SE$

1.2 Confidence interval of mean

- Sample Mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
- Standard Error: $SE = \frac{s}{\sqrt{n}}$
- Confidence Interval: $CI = \bar{x} \pm z \cdot SE$

1.3 CI Overlap Rules of Thumb (Heuristic)

- 1.3.1 Conditions
- One of the CI must be **control group**.
 - Both groups must be **independent** (RCT or equivalent, no pairing).
 - Same variable, same conditions
- 1.3.2 Rules of thumb:

Let two 95% confidence intervals be $[L_1, U_1]$ and $[L_2, U_2]$. Overlap = Overlap in CIs.

- No overlap -> difference is statistically significant at the 95% confidence level.
- Overlap $< \frac{1}{3} \times$ range covered by the CIs -> could be statistically significant. Bigger Overlap -> less confidence.
- Overlap $> \frac{1}{3} \times$ range covered by the CIs -> difference is probably not statistically significant.

Note: This is a heuristic approximation, to be used because HSI taught this.

2 Lecture 1: The Founding of Modern Science

- Science emerged as a formal method during the Scientific Revolution (1543 De Revolutionibus → 1687 Principia).
- Key figures (Copernicus, Galileo, Kepler, Bacon, Newton) shifted from Aristotelian views to evidence-based understanding.
- The steam engine and Industrial Revolution show how discoveries enabled technological and societal transformation.

3 Lecture 2: The Baloney Toolkit Applied to a Simple Investigation

- Source reliability: credentials · lateral reading
 - Perspective: funding/purpose → bias
 - Positive evidence: supports · reliable · relevant · accurate data
 - Evidence consensus: majority viewpoint
 - Verified independently: reproduction · peer-review · fact-check
 - Sound reasoning: no fallacies · correct logic
- Prep: skeptical mindset · aware of own biases · guard emotions

4 Lecture 3: Scientific Explanations and Models

- Explanations must account for observations and be falsifiable (capable of being proven wrong).
- Models are simplified representations that help understand complex systems.
- Understand models' limitations and assumptions before applying them to real situations.

5 Lecture 4: Experimentation and Uncertainty

- All measurements have inherent uncertainty; precision indicates reliable significant figures.
- Accuracy compares the mean to the true value; uncertainty in averages is expressed as margin of error (commonly 95

- Validate and calibrate instruments/methods to remove systematic errors and ensure true measurements.

6 Lecture 5: The Science of Climate Change

6.1 Climate Change Myths and Truths

1. Myth: Climate's Changed Before

- Natural cycles → current warming natural
- ✓ Milankovitch cycles, solar radiation: exist
- ✓ Current warming: unprecedented (rate & magnitude, 170 yrs)
- ✓ Driver: human GHG ≠ natural variation

2. Myth: It's the Sun

- × Solar activity → warming
- ✓ Past 40 yrs: solar ↓ slight, temps ↑ significant
- ✓ Pattern (strat cool, tropics warm): GHG forcing ≠ solar
- ✓ Sun: not primary driver

3. Myth: Animals & Plants Adapt

- × Flora/fauna → adapt
- ✓ Extinction: 100–1000× baseline
- ✓ Pace too rapid: no adaptation time
- ✓ Outcome: species extinction

4. Myth: Earth Cooling

- × Warming stopped, cooling now
- ✓ Cherry-picking time windows: false
- ✓ Full dataset: ↑ unambiguous
- ✓ 2010s: warmest decade

5. Myth: It's Not Us

- × Climate change ≠ human
- ✓ 10 fingerprints:
 - CO₂: 30 Gt/yr ↑
 - O₂: ↓
 - Fossil fuel carbon signature
 - Coral carbon absorption
 - Heat escape: specific wavelengths
 - Atmosphere: shrinking
 - Nights: warming faster
 - Troposphere: warm, stratosphere: cool
 - Ocean heat: ↑
 - Fossil carbon: ↑ in air

Global Warming Evidence

- All indicators ↑ (as expected GHG):
 - Land surface temps
 - Sea surface temps
 - Ocean heat (90% of total)
 - Sea level
 - Glacier melt (Arctic ice ↓ 35% since '79)
 - Snow cover ↓
 - Humidity ↑

Five Denial Forms

- Science: "no consensus"
- Economic: "too costly"
- Humanitarian: "net benefit"
- Political: "others act first"
- Crisis: "uncertainty → inaction"

Cognitive Biases

- Time-discount: future ↓ valued
- Optimism: "tech solves it"
- Control illusion: "scientists fix"
- Egocentrism: "others blame"

Atmospheric Mechanisms of Climate Change

Atmospheric Composition & Greenhouse Gases

- Major gases: N₂ (78%), O₂ (21%), Ar (1%)
- Trace gases (greenhouse effect):
 - CO₂: ~420 ppm (pre-industrial: 280 ppm)
 - CH₄: ~1900 ppb
 - N₂O: ~335 ppb
- Greenhouse gases → absorb infrared radiation → trap heat

Infrared Radiation & Energy Balance

- Solar constant: $S = 1370 \text{ W m}^{-2}$
- Earth cross-section: $\pi R^2 \rightarrow$ absorbed energy: $S \times \pi R^2 / 4 \pi R^2 = S/4$
- Stefan-Boltzmann law: $P = \sigma T^4$ (emitted energy $\propto T^4$)
- IR wavelengths: 12–18 μm (thermal radiation from Earth)
- CO₂ absorption: 90× more effective than air
- H₂O absorption: 16,000× more effective than air

- Atmosphere: transparent to incoming solar radiation ↔ opaque to outgoing IR

Temperature & Greenhouse Effect

- Without atmosphere: $T_E \approx -18 \text{ }^\circ\text{C}$ (calculated from energy balance)
- With atmosphere (greenhouse effect): $T_E \approx +15 \text{ }^\circ\text{C}$ (observed)
- Difference: $\sim 33 \text{ }^\circ\text{C}$ warming due to greenhouse gases
- Mechanism:
 - Atmosphere absorbs outgoing IR radiation
 - Atmosphere re-emits radiation in all directions (including back to surface)
 - Additional downward radiation → surface warming
- CO₂ increase → more IR absorption → higher surface temperature

Carbon Dioxide & Causal Chains

- Fossil fuel combustion → CO₂ emissions → atmospheric CO₂ ↑
- Rising CO₂ (280 → 420 ppm since 1750):
 - Due to: coal, oil, natural gas burning
 - Confirmed by: carbon-14 isotope ratios (fossil fuels lack ¹⁴C)
- CO₂ increase → IR absorption ↑ → temperature ↑
- Temperature rise unprecedented in past 800,000 years (ice core records)

Troposphere & Stratosphere

- Troposphere (lower atmosphere):
 - Height: 0–12 km (varies: ~8 km poles, ~18 km equator)
 - Temperature: decreases with altitude (~6.5 °C per km)
 - Contains: weather systems, clouds, 80% of atmosphere's mass
 - Greenhouse effect warming occurs here
- Stratosphere (upper atmosphere):
 - Height: 12–50 km
 - Temperature: increases with altitude (inversion due to ozone)
 - Contains: ozone layer (~20–25 km altitude)
- Boundary (tropopause): temperature minimum (~–57 °C)

UV Radiation & Stratospheric Processes

- Solar UV radiation → reaches upper atmosphere
- Ozone formation: O₂ (UV) → O → O₃ (ozone)
- Ozone role: absorbs UV radiation → protects surface → heats stratosphere
- Temperature structure:
 - Troposphere (cools with altitude): due to greenhouse effect dominance
 - Stratosphere (warms with altitude): due to UV + ozone heating
- CFCs → ozone depletion → less UV absorption → more surface UV

What Affects What: Causal Mechanisms

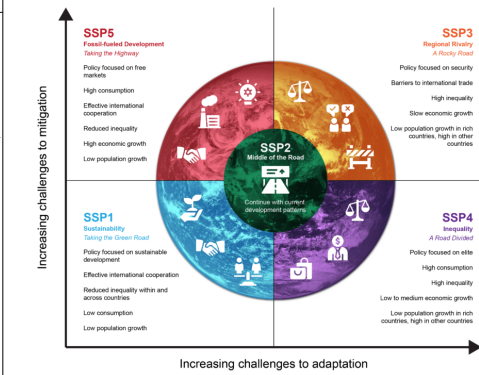
Fossil fuels → CO₂ ↑ → IR absorption ↑ → Temperature ↑

- Solar input (~1370 W/m²) → relatively constant
- CO₂ concentration → controls fraction of IR trapped
- Atmospheric composition → determines temperature equilibrium
- Temperature → drives all climate effects (precipitation, sea level, ecosystems)

Energy Balance Model (Summary)

- Energy in: Solar radiation absorbed = $S \times (1 - A) \times \pi R^2 / 4$
 - S: solar constant
 - A: albedo (reflectivity ~ 0.3)
- Energy out: IR radiation emitted = $\sigma T_E^4 \times 4\pi R^2$
- Equilibrium: Energy in = Energy out → Solve for T_E
- Greenhouse effect: reduce effective emissivity → higher T_E at equilibrium
- More CO₂ → lower effective emissivity → higher T_E

7 Lecture 6: Modelling Climate Change



SSP 1: Scientific Consensus History

- Describe establishment of scientific consensus on climate change
- Include role of IPCC: Intergovernmental Panel on Climate Change
- Track: 1979 First Assessment → 1995 Second Assessment (discernible impact) → 2001 Third Assessment (mostly due to humans)

SSP 2: Climate vs Weather & Greenhouse Effect vs Climate Change

- Climate: average weather over long period in a region
- Weather: short-term atmospheric conditions → temporal scale: days
- Greenhouse effect: GHGs trap IR radiation → warming (natural process)
- Climate change: shift in average climate conditions → temporal scale: decades/centuries
- Distinction: global warming (cause) ⇒ climate change (effect)

SSP 3: Proxy Methods for Past Temperature Reconstruction

- Dendroclimatology: tree ring width $\propto$ growth conditions ⇒ historical climate estimate
- Coral records: oxygen isotope ratios in carbonate shells $\propto$ temperature
- Ice cores: composition of trapped air bubbles ⇒ atmospheric conditions; oxygen isotopes $\propto$ temperature
- Paleoclimate proxies: calibrated against modern measurements ⇒ extend record backwards
- Data: 800,000 years available; recent warming unprecedented in last 2,000 years

SSP 4: Public vs Scientific Consensus Gap

- Scientific consensus: 97.1% of peer-reviewed papers agree on anthropogenic climate change
- Public perception: 1997 only 48% believed consensus → 2021 rose to 68% (still below actual)
- Disconnect: public underestimates scientist agreement on climate change reality
- Oreskes (2004): 928 papers surveyed, 100% agreed; 0% rejected consensus

SSP 5: Establishment of Scientific Consensus

- Evidence requirements: (1) observational data showing warming, (2) temperature rise unusual, (3) GHG increases, (4) CO₂ rise from fossil fuels
- Key evidence:
 - Instrumental record: NASA, NOAA, Met Office all show ~1.1°C rise since pre-industrial
 - Proxy records: trees, coral, ice cores ⇒ current warming unprecedented in 800,000 years
 - CO₂ rise: 315 ppm (1958) → 415+ ppm (today); Keeling curve shows continuous increase
 - Carbon-14 effect (Suess): fossil fuels dilute atmospheric ¹⁴C ⇒ directly proves fossil fuel origin
- Verification: multiple independent datasets, methods, organizations all converge ⇒ consensus achieved

• Consensus definition: majority of scientists agree based on verified evidence, not authority 8 Lecture 7: On the Mongering of Doubt • Scientific uncertainty is sometimes misrepresented as disagreement to cast doubt on established findings. • Well-funded campaigns manufacture doubt using tactics similar to those of the tobacco industry. • Media and political polarization amplify manufactured doubt despite overwhelming evidence. 9 Lecture 8: The Reliability of Climate Change Predictions • Climate predictions compare well with observed data and have proven reliable in many cases. • Individual models vary; ensemble and trend analyses show consistency with observations. • Confidence arises from multiple independent lines of evidence converging on similar conclusions. 10 Lecture 9: Restoration of Nature (Sample Science) • Restoration ecology applies scientific methods to return degraded ecosystems to functional, biodiverse states. • Understanding ecological processes and species interactions is essential for successful restoration. • Long-term monitoring and adaptive management allow strategy adjustments based on outcomes. 11 Lecture 10: Connectivity of Nature (Sample Science) • Ecological connectivity—movement between habitat patches—is critical for biodiversity conservation. • Fragmentation reduces connectivity, isolating populations and lowering genetic diversity and resilience. • Corridors and connected protected-area networks enable movement and maintain ecosystem function. 12 Lecture 11: Hinterland of Resources (Sample Science) • Resource hinterlands that supply cities often suffer significant environmental degradation. • Understanding resource flows and ecological costs is key for sustainable management. • Local and regional policies shape extraction practices and environmental impacts. 13 Lecture 12: Fallacies in the Name of Science & Course Wrap Up • Common logical fallacies (appeal to authority, straw man, ad hominem) misrepresent scientific claims. • Understanding science, its limits, and proper evidence evaluation guards against pseudoscience. • Science is powerful for understanding the natural world but must be applied critically, aware of uncertainty and evidence. 13.1 Fallacies & Cognitive Biases • False Anomaly: Presenting an expected or documented phenomenon as mysterious, then invoking extraordinary explanations while ignoring simpler, documented causes (e.g., crop circles blamed on aliens while ignoring tractor tramlines). • Questionable Arguments by Elimination: Ruling out some alternatives does not prove the remaining hypothesis (eliminating luck doesn't prove ESP). • Illicit Causal Inferences (Post hoc): Inferring causation from correlation or temporal proximity (thinking of a friend then they call ≠ your thought caused the call). • Unsupported Analogies: Using analogies or superficial similarities as evidence when they do not justify the inference (e.g., moon→tides so planets→personality). • Untestable Explanations & Predictions: Claims that cannot be empirically falsified are unscientific; conspiracy theories often evade falsification by treating absence of evidence as "proof" of a cover-up. • Empty Jargon: Stringing together scientific-sounding but meaningless terms to create an appearance of expertise (e.g., "quantum entanglement telekinesis"). • Ad Hoc Rescues: Continuously modifying a claim to accommodate contrary evidence instead of revising or abandoning the claim. • Exploiting Uncertainty: Weaponizing legitimate scientific uncertainty about details to cast doubt on well-established conclusions (e.g., using model uncertainty to deny consensus on climate <td data-bbox="560 0 1120 1596"> change). • Confirmation Bias / Cherry-picking: Selecting evidence that supports a preferred view while ignoring contradictory data. • Appeal to Authority / False Authority: Accepting a claim solely because an alleged authority says so, especially when outside their expertise. • Straw Man: Misrepresenting an opponent's argument to make it easier to attack. • Anecdotal Evidence: Treating isolated stories or anecdotes as proof against systematic data. • False Equivalence: Presenting two unequal arguments or bodies of evidence as if they were equivalent. 13.2 Cognitive Profile: Dunning-Kruger Effect • Low-ability individuals tend to overestimate their competence; conversely, highly competent people may underestimate theirs. • Awareness of this effect helps interpret confident claims from non-experts and cultivates intellectual humility. 13.3 Good Practices & Non-fallacies • Healthy skepticism: Question claims constructively and demand appropriate evidence before accepting extraordinary conclusions. • Burden of proof: Assign stronger evidential requirements to more extraordinary claims; require reproducible, falsifiable, and preferably independent evidence. • Consider alternative hypotheses, actively seek disconfirming evidence, and weigh converging independent lines of evidence. </td>	change). • Confirmation Bias / Cherry-picking: Selecting evidence that supports a preferred view while ignoring contradictory data. • Appeal to Authority / False Authority: Accepting a claim solely because an alleged authority says so, especially when outside their expertise. • Straw Man: Misrepresenting an opponent's argument to make it easier to attack. • Anecdotal Evidence: Treating isolated stories or anecdotes as proof against systematic data. • False Equivalence: Presenting two unequal arguments or bodies of evidence as if they were equivalent. 13.2 Cognitive Profile: Dunning-Kruger Effect • Low-ability individuals tend to overestimate their competence; 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